Towards Reconstructing the Brain via Graph Signal Processing

April 4, 2024

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***Abstract*—Graph Signal Processing (GSP) presents a novel method to analyse data. This paper, via a literature review [1], presents the applications of GSP in neuroscience. Specifcally, it explores *why graph signal processing plays a crucial role in understanding the structural and functional connections in the brain.* Classical brain imaging techniques and how GSP tools can be used to mitigate their shortfalls are first presented. Two applications of GSP in neuroscience are then probed. One, understanding the structural and functional connectivity of the brain, and two, diagnosing mental disorders using graph theory and GSP. Finally, the limitations of using GSP in the applications above are critiqued by juxtaposing popular deep-learning techniques with GSP. Currently, GSP alone cannot replace the former but can be used in conjunction with deep-learning, by using GSP for dimensionality reduction, or for generating connectivity matrices as input for models.**

***Index Terms*—Brain connectivity, graph signal processing, neurological disease diagnosis, neuroscience**

I. Introduction

Classical data analysis has been carried out in simple domains- time, or space. These "regular" domains are easy to visualize intuitively. However, data often lives in highly irregular, complicated domains. Graphs can be used to capture these complex data structures [1]. Social networks are one example. Nodes on a graph can represent individuals, with their connections represented by edges. Classical signal processing is a mature field, with well-understood methods. Representing data on graphs allows us to port these mature techniques onto this irregular network. Viewing data as signals on a graph can also reveal interesting properties about the behavioural dependence between nodes. One such domain where the application of graph signal processing (GSP) has uncovered new revelations is neuroscience, which will be the subject of this review paper.

The study of patterns in the brain has proven essential in understanding neurological diseases. Analysis of brain data has been performed on several data types. Time series data, Electroencephalograms (EEGs) and image data- functional Magnetic Resonance Imaging (fMRI), have been popular subjects of analysis to diagnose neurological diseases and provide insights into how the brain functions. However, interpreting brain data as a signal on brain connectivity graphs reveals patterns of interest unavailable through conventional means [2].

One such example of a GSP tool that can be used to understand underlying brain structure is network inference [3]. Network inference is a tool to understand topology in data without directly observable structure. Utilizing these tools in the context of brain data can provide insights into the behavioural connections between different parts of the brain. Understanding brain connectivity can provide further insight into neurological disease progression. Abnormal brain signals can be captured between functional brain regions identified by network inference tools in GSP. Disorders such as autism [4], Alzheimer's [5] and schizophrenia [6] can be better understood, allowing for early detection and treatment.

GSP tools can thus be used to study the brain from a different perspective- gaining insights into how the brain works, and how diseases present themselves. This paper explores *why graph signal processing plays a crucial role in understanding the structural and functional connections (SC, FC) in the brain.* The shortfalls of classical brain-imaging techniques, and how they can be overcome with GSP are first presented. Applications of GSP in understanding brain connectivity and disease diagnosis are then discussed. The paper concludes with future research directions by juxtaposing popular deep-learning techniques and the GSP tools discussed.

II. Methodology

The methodology followed to write this review article first involved using a highly cited GSP review paper as a base [1]. The applications of GSP in neuroscience were then selected to be explored specifically. Using Connected Papers and articles about the domain cited in the base paper, several publications were shortlisted for analysis. Each paper's relevance was then evaluated using Google Scholar by reviewing their abstract, introduction and results. The expert opinions of Alberto Natali and Ruben Wijnands, PhD students in the Signals and Systems group were also taken into account to select highly relevant papers.

[1]**This literature review was written at Delft University of Technology for the course EE4C01 Profile Orientation and Academic Skills, with the collaboration of Drs. Regina J. Hoffmann.**

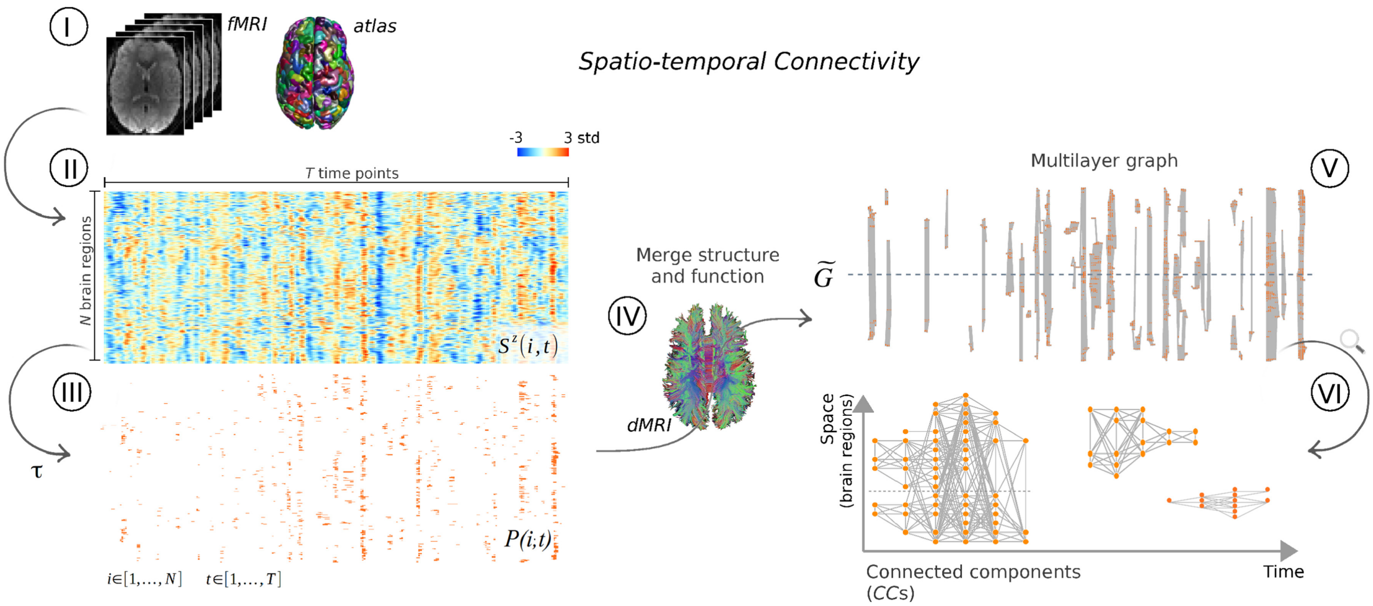

Fig. 1. Construction of a spatio-temporal connectome: (I) fMRI data collected and segmented to N brain regions, with each region having an average functional time series length of T time points. (II) Regional fMRI data given a z-score and presented as a matrix $S^z(i,t)$ (III) Reduced to binary point process $P(i,t)$ by applying a threshold τ. (IV) Structural connectivity information derived from dMRI tractography (V) Combining (III) and (IV) to create the multilayer graph $\tilde{G}$ (IV) zoomed version of the connected components in $\tilde{G}$ [7]

III. Classical Methods of Brain Analysis

Brain Imaging techniques have helped the rapid development of cognitive neuroscience [8] by allowing researchers to observe brain activity directly while subjects perform tasks. This section describes their utility to researchers, their pitfalls, and how GSP could potentially be used to help overcome these. One example of a traditional imaging technique is diffusion Magnetic Resonance Imaging (d-MRI). d-MRIs track the movement of water molecules within human tissue unveiling the brain's SC. They particularly uncover the microarchitecture of axon fibre tracts in the brain [9]. This structural information can help diagnose neurological disorders including multiple sclerosis, dementia, and strokes [10]. Another example is Resting-State Functional MRIs(rs-fMRI). f-MRIs provide insights into how different parts of the brain interact during a task by measuring metabolic changes [8], in this case during rest. As opposed to d-MRIs, f-MRIs provide information regarding functional connectivity. Several shortfalls of traditional methods have been described in [7]. The authors explain that there has been an absence of research that combines the two- "how functional interactions unfold through the structural connectome" has been seldom studied. Traditional methods also assume that during rest, brain activity is stationary. Lately, this assumption has been proven false through several studies that have attempted to study resting-state dynamics- it has been shown that functional patterns in the brain can change in short time scales, instead of being stationary. The question then is, how different sources of vital information about the brain could be combined to yield better results in disease diagnoses and to better understand how the brain works?

As expressed by A. Griffa and her colleagues [7] the functional interactions that take place in the brain can be captured by using GSP. Using a multi-layer graph that can show the physical structural connections and follow the time-varying functional connections, it can unveil the communications channels, which would otherwise not be visible from the MRIs alone. The SC graph is constructed using dMRI tractography which expresses the adjacency in the different brain regions. In contrast, the temporal changes are detected by using a point process analysis, like thresholding, of the rs-fMRI to show the activation of different regions at neighbouring time points. By combining them both and creating the "spatio-temporal connectome framework", the activity pathways can be measured and reproducible spatio-temporal patterns can be identified as shown in Figure 1. This can be used to understand the communication mechanisms and functional dynamics within the brain.

GSP can also be used for dimensionality reduction for classification problems. M. Menoret and his team [11] found that by combining the geometrical structure and functional connectivity and decomposing these signals using the Graph Fourier Transform, the spectral components of the brain signals are obtained. This can be done as the eigenvectors of the graph matrix have a similar Fourier mode. After this graph sampling is done to reduce the dimensionality and this method is proven to yield better classification results compared to classical methods like Principal Component Analysis and Independent Component Analysis.

IV. Applications

Graphs and networks can be used to structure relationships between pairs of elements in a set. This simple explanation allows GSP to be used in various disciplines that range from biology, medicine, and psychology to sociology, economics, and engineering [3]. This section details the use of GSP in brain imaging applications; namely, its use in understanding SC and FC in the brain and diagnosing medical conditions such as autism, schizophrenia, and Alzheimer's.

A. Understanding Connectivity

In the brain, "connectivity" refers to how different parts of the brain communicate with each other. It presents itself as structural, functional, and effective [12] connectivity. GSP can model SC by representing the brain's anatomical connections as a graph where the nodes correspond to different brain regions and the edges correspond to the physical connections between these regions such as white matter tracts [11]. The FC is represented with GSP by using the statistical dependencies between brain signals [13].

In neuroscience, task repetition refers to the practice of performing the same cognitive or physical task multiple times. Researchers can study how brain activity patterns change over time with repeated fMRI scans during the learning and recovery processes [11]. By applying graph frequency analysis to this fMRI data, researchers can enhance their understanding of how brain networks adapt themselves and reorganize during the learning process of task repetition [2]. With graph frequency analysis, spatial variations in the brain signal can be studied; low graph frequencies represent smooth and regular variations in the brain, whereas high graph frequencies represent important spatial variations [11].

B. Diagnosing Medical Conditions

Modern graph theory and GSP can be used to diagnose mental disorders such as autism, Alzheimer's and schizophrenia [4] [5] [6]. Current approaches to diagnosis include generally include clinical assessments and behavioral observations which can increase the risk of misdiagnosis or missed diagnoses [14]. By relying on quantitative metrics, a more objective diagnosis can be made with potentially lower false positive and missed diagnosis rates. Further, studying the brain abnormalities which result from these diseases can offer significant insights into the brain help uncover the underlying causes of the disease.

When detecting presence of autism in the underlying brain network, the brain exhibits over-connectedness overall, a higher betweenness, a higher global efficiency but lower local efficiency, and an overall shorter path length [4]. These conclusions are medically consistent with known information regarding the development of Autism. The current hypothesis for autism diagnosis indicates a lack of connection pruning during adolescence causes reduced network segregation and longer switching times. This manifests itself behaviorally with slower adjustments to environmental changes. This is in accordance with an observed trait of autistic people – resistance to change [4].

The aforementioned graph metric results have been deduced from fMRI data. The researchers assume the same underlying node placement across patients and focus on identifying presence or absence of connections with the fMRI data. This analysis suggests autism diagnosis is indeed possible by examining known graph evaluation metrics.

A similar approach to diagnosis has been applied to brains with Alzheimer's using positron emission tomography (PET) imaging data [5]. Different to the autism diagnosis, however, Hu and colleagues propose Graph Regression Models (GRM) to learn a patient's specific underlying graph structure. Regions of interest are identified based on similar intensity values, forming nodes. Nodes are assigned values based on a corresponding the intensity of the PET image forming a graph signal. An assumption of a smooth graph signal is additionally imposed. With knowledge of nodes, the connections between can be estimated by solving an optimization problem where the graph structure best explains the observed PET signals given a particular graph signal smoothness. Parameters such as smoothness and sparsity are fine tuned based on expert knowledge [5]. The GRM creates a patient specific graph for analysis with known graph theory metrics. Results suggest the reconstructed network is more accurate than existing methods implying that the disease diagnosis which follows is also more accurate.

Graph theory metrics remain tried and true in recognizing abnormalities in Alzheimer brains. These brains exhibit a lower overall clustering coefficient, a lower global efficiency, and less modularity within specific functional regions. This verifies the use of graph based approaches for Alzheimer diagnosis.

Song and colleagues approach disease diagnosis with fMRI data from a different perspective [6]. Schizophrenic brains show increased and strengthened connections in various functional regions hypothesized as potential causes for hallucinations and delusions. Graph signal smoothness is used to identify schizophrenia by analyzing brain activity consistency across patients. Patients are nodes in a graph and the similarity of their brain activity link these nodes. A lack of smoothness indicates the patient is more likely to be schizophrenic providing a classifier for medical professionals.

Disease diagnosis is certainly possible with graph-based approaches. Classical approaches use functional regions as nodes and use graph theory metrics to identify affected patients [5]. Other groups choose to generate user-specific graphs as seen with the GRM prior to performing a similar graph theory based evaluation [4]. Lastly, patients can themselves be represented as nodes in a graph building a classifier for disease diagnosis. The aforementioned approaches for disease diagnosis demonstrate the wide breadth of potential for graph theory and GSP in this domain.

V. Discussion

In this section, the application of GSP in understanding brain connectivity and medical diagnosis will be discussed from a critical point of view. The limitations of GSP, prevailing arguments against these, and a rebuttal to present the potential and adaptability of GSP for neuroscience will be given.

A. Application to Understanding Brain Connectivity

The application of GSP in understanding SC, and FC of the brain has great potential in neuroscience studies, especially in imaging applications done with fMRI. However, GSP has some undeniable limitations.

In the case of SC and FC, the precision of GSP and the results that it provides rely heavily on the quality and resolution of the data. Misinterpretation of the data or inaccuracies in the data might lead to false interpretations of connectivity and this impacts the efficacy of GSP in understanding brain structures and functions. Another one of GSP's disadvantages comes from its reliance on accurate graph structures. The effectiveness of GSP hinges on the premise that the nodes correspond to brain regions and the edges correspond to the physical connections, on the graph. This premise might not always be accurate due to the dynamic and complex nature of the brain and brain networks leading to an oversimplification. Also, most GSP graph models assume linearity. Due to this assumption, GSP may be adequate in capturing the complexities of brain connectivity which often exhibits non-linear characteristics.

While there is extensive research related to GSP's applications on SC and FC, effective connectivity poses a more complex challenge due to its focus on the causal effects of the brain signals and directional influence between brain regions [12].

Despite these challenges, GSP still remains efficient in understanding SC, and FC of the brain. Studies by Huang and his colleagues [2] in the context of functional brain imaging show promising results in this regard about the attention switching problem. Also, the problem of graph structure representation is not unique to GSP and it is a common challenge across all computational models in neuroscience. Developments in machine learning can offer robust methodologies for enhancing graph structure accuracy and help the development of adaptive and dynamic graph models which can also address the concerns regarding non-linear and evolving brain connectives.

B. Application to Medical Diagnosis

Using graphs, brain imaging data can be a quantitative method of disease diagnosis. The classical approach involves defining node connections via fMRI data and applying graph theory as a metric for disease diagnosis [5]. Proven and reliable, this method is the first step in using graphs to analyze brains. However, the results are not yet widely studied enough across the global population to be used as a diagnostic tool. Further, the differences between normal and abnormal brains are vague with statements of "lesser" connections or "more" rather than quantifiable measures.

In an attempt to improve upon the approach, alternative methods of graph formulation have been proposed such as the GRM [4]. The GRM, however, produces a graph which is still very similar to the classical graph formulation approach as described in [5]. The minimal change in the underlying graph limits the possible improvements for disease diagnosis.

Taking an alternative approach by representing patients themselves as nodes in a graph, Song et al. builds a classifier based on graph signal smoothness, a GSP concept. The results, albeit promising, are preliminary and only "generally consistent" with the known data. [6]

These studies provide a promising first step in demonstrating the feasibility of medical disease diagnosis using graph theory and GSP. They confirm brains have an underlying graph structure. Further, they verify disease detection is possible from brain structure alone. However, these approaches do not fully utilize the potential of the given data. While a focus was placed on fMRI data, other imaging techniques such as reveal how brain signals navigate across regions and time. When paired with the known graph structure of diseased patients and GSP, more powerful disease detection techniques can be discovered. Further, the movement of the signals can provide insights into the diseases themselves.

C. Future Work

GSP does have the potential to be paired with different imaging modalities and techniques to yield a clear analysis of the brain's connectome. As mentioned earlier, GSP cannot outperform Deep learning models currently but can be combined with them whether it be for dimensionality reduction [11] or using the connectivity matrix generated as input for models [14]. Many studies using graph theory have been done to aid in Deep Brain Stimulation for various conditions like Parkinson's [15] [16] and epilepsy [17]. By utilizing the ability of GSP to generate the connectivity graph, these results could be improved and translated to other conditions as well. Not only for imaging modalities, but GSP can be used to analyze electroencephalography data [18] and even combining multiple modalities could lead to a new perspective of analyzing brain data.

VI. Conclusion

Understanding the brain is a complex task which can be made simpler by the use of graph signal processing (GSP). The brain's connectivity within regions and the diagnosis of medical conditions are two real world applications of interest. Existing literature is explained, limitations are discussed and areas of future work are identified. While the limitations are vast, the current studies demonstrate a feasible future for GSP in unraveling the hidden mysteries of the brain.

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